Burhan Ahmad Wani

Engineering, GIS, Spatial Analysis •+44-7774685957 • London, UK

Executive Summary

Dynamic professional at the intersection of Urban Planning, Spatial Data Science, Transport Studies, and Civil Engineering, with a Master's in Urban Spatial Science from UCL. Demonstrated expertise in applying advanced spatial analytics, sensor data, machine learning, and satellite technologies to urban environments. Extensive experience in large-scale urban projects, including the £15 million National Road Program, and academic research in urban heat island mapping, built environment analysis, and strategic transport planning and committed to leveraging data and technology for sustainable, resilient, and technology-driven urban development aligned with the UN SDGs.

Educational Information

University College London (UCL) London, UK

- MSc, Urban Spatial Science. DISTINCTION
- Commonwealth Scholar -23/24 (
- Bartlett Student Ambassador 23/24

National Institute of Technology (NIT) Srinagar, India

- Bachelor of Technology (BTech), Civil Engineering. GPA 4.0
- Silver Medalist with a CGPA of 9.241/10 among 120 students.

Experience

Public Works Department (JKPWD), Govt. of India India
Assistant Engineer, PMGSY Mar 2017 – Aug 2023

- Led the implementation of the Indian National Rural Road Plan (PMGSY-III) to connect rural populations in Northern Region of India, managing spatial planning, groundwork, & implementation.
- Ensured effective communication within the team & served as a liaison between multiple stakeholders.
- Recognized as "Outstanding Officer" by the Government of J&K, India for 2021-22.

Projects

- **Urban Heat Island Dynamics in London Wards: The Role of Built Environment Features:** Investigated the impact of built environment characteristics on the Urban Heat Index using data from the London Datastore and Google Earth Engine. Employed multiple linear regression to analyze how various features influence UHI across different London wards.
- **Strategic Route Planning and Economic Analysis for Delivery Services of a City:** Developed a NetLogo model to optimize operational strategies and financial performance using real-world geographic data.

- **Crop Yield Estimator:** Conducted a detailed analysis of crop yield using historical data and machine learning algorithms. Employed techniques such as linear regression, decision trees, and random forests to predict crop yield and identify key influencing factors.
- **Impact of Built Environment Factors on House Prices in London Boroughs:** Investigated the impact of built environment characteristics on average House Prices using data from the London Datastore. Employed multiple linear regression to analyze how various features influence House Prices across London boroughs.
- **Analyzing Spatial Clustering of Public Service Facilities in Andhra Pradesh, India:** Explored the spatial clustering of public service facilities post-bifurcation using K-means, Hierarchical, and DBSCAN clustering methods. Analyzed the distribution of health, transport, education, and agricultural services at the block level, identifying regional disparities and service density variations.

Achievements

- Selected for three prestigious Russell Group Universities in the UK with full scholarships: Tetley and Lupton Scholarship (2023), International Masters Excellence Scholarship (2023), FELS - Southampton Environmental & Life Sciences Dean's International Scholarship, and Commonwealth Scholarship (2023-24).
- Awarded Science and Technology Merit Scholarship by the Dept. of Science & Technology, J&K (2006-2008).
- Received Certificate of Merit by British Safety Council (2017) for commitment to high standards of health, safety, and environment.
- Received Certificate of Appreciation by Govt. of J&K for leadership and effective communication with the Fire Department during an off-site fire and emergency incident.

Skills & Affiliations

- **Skills:**
  - **Programming Skills:** Python (pandas, NumPy, GeoPandas, scikit-learn, TensorFlow, Keras, matplotlib, seaborn, statsmodels), R (tidyverse, sf, tmap, janitor, spatstat, stringr, sp, RColorBrewer, tmaptools, OpenStreetMap, readxl, spdep, viridis), Quarto, NetLogo, Xaringan, Latex (Overleaf), Git.
  - **Geospatial Tools:** QGIS, Google Earth Engine (GEE).
  - **Machine Learning & Data Analysis:** Artificial Neural Networks (ANNs), Gradient Boosting (XGBoost), Random Forest, Linear Regression, K-means, Hierarchical Clustering, DBSCAN, Data Cleaning, Feature Engineering, Visualization, Statistical Analysis.
  - **Data Management and Manipulation:** Big Data, Data standardization, missing data imputation, geospatial analysis, time-series analysis.
  - **Additional Skills:** Microsoft Office Suite (Word, Excel, PowerPoint), Google Workspace (Docs, Sheets, Slides, Forms), Design and Visualization (SketchUp Pro, Google Earth Pro, Revit, Canva, Tableau, Draw.io), Slack, etc
- **Professional Memberships:**
  - Engineering for Change (E4C); Institution of Engineers of India (IEI), Institution of Civil Engineers (ICE)
- **Volunteering:**
  - Indian Red Cross Society, Volunteer Centre Camden & Islington, London, UK.